CHAPTER 9

STRAIGHT LINES

CHAPTER ESSENCE AT A GLANCE:

Geometry meets algebra! Every straight line on a graph is described by a simple linear equation. Slope m is simply rise over run. Whether you know two points, a point and slope, or two axis intercepts, you can write the line equation in under 30 seconds!

4 CORE PRINCIPLES TO UNDERSTAND FIRST

- **Slope of a Line:** m = (y_2 - y_1) over (x_2 - x_1) = tan θ. Parallel lines have m_1 = m_2. Perpendicular lines have m_1 · m_2 = -1 (m_2 = -1 over m_1).
- **Standard Equation Forms:** Point-Slope: y - y_1 = m(x - x_1) ; Slope-Intercept: y = mx + c ; Intercept Form: x over a + y over b = 1 (a = x-intercept, b = y-intercept).
- **Acute Angle Between Two Lines:** tan θ = |(m_2 - m_1) over (1 + m_1 m_2)| (use absolute value for acute angle).
- **Perpendicular Distance from Point:** Distance from (x_1, y_1) to Ax + By + C = 0 is d = |A x_1 + B y_1 + C| over sqrt(A^2 + B^2).

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \tan\theta, \quad L_1 \perp L_2 \iff m_1 m_2 = -1$$

Slope Definition and Orthogonal Lines Criterion

$$y - y_1 = m(x - x_1) \quad \text{(Point-Slope)}, \quad y = mx + c \quad \text{(Slope-Intercept)}$$

Standard Analytical Forms of Straight Line Equations

$$\frac{x}{a} + \frac{y}{b} = 1 \quad (\text{Intercept Form: } x\text{-int} = a,\ y\text{-int} = b)$$

Symmetric Intercept Form of Straight Line

$$d = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$$

Perpendicular Distance of Point (x1, y1) from Line Ax + By + C = 0

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

- **Alert:** Parallel lines have equal slopes (m_1 = m_2). Perpendicular lines have product of slopes m_1 · m_2 = -1 (m_2 = -1 over m_1).
- **Alert:** The angle θ between two lines is given by tan θ = |(m_2 - m_1) over (1 + m_1 m_2)|. Always take the positive absolute value for acute angle!

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Find the equation of the line passing through (-3, 5) with slope -2. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Point-Slope Form]** y - y1 = m (x - x1). Given (x1, y1) = (-3, 5) and m = -2. [1 Mark]

• **[Step 2: Simplification]** y - 5 = -2 [x - (-3)] = -2 (x + 3) = -2x - 6 ⇒ 2x + y - 5 + 6 = 0 ⇒ 2x + y + 1 = 0. [1 Mark]

• **[Boxed Result: 2x + y + 1 = 0]**

2. Find the equation of the line passing through the points (1, -1) and (3, 5). [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Two-Point Form]** y - y1 = [(y2 - y1) over (x2 - x1)] (x - x1). Slope m = [5 - (-1)] over (3 - 1) = 6 over 2 = 3. [1 Mark]

• **[Step 2: Equation]** y - (-1) = 3 (x - 1) ⇒ y + 1 = 3x - 3 ⇒ 3x - y - 4 = 0. [1 Mark]

• **[Boxed Result: 3x - y - 4 = 0]**

3. Find the angle between the lines y - sqrt(3) x - 5 = 0 and sqrt(3) y - x + 6 = 0. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Slopes of Both Lines]** Line 1: y = √3 x + 5 ⇒ m1 = √3. Line 2: √3 y = x - 6 ⇒ y = (1 over √3) x - 6 over √3 ⇒ m2 = 1 over √3. [1 Mark]

• **[Step 2: Angle Formula]** tan θ = |(m2 - m1) over (1 + m1 m2)| = |[1 over √3 - √3] over [1 + √3(1 over √3)]|. [1 Mark]

• **[Step 3: Calculation]** Numerator = (1 - 3) over √3 = -2 over √3. Denominator = 1 + 1 = 2. tan θ = |-2 over (2 √3)| = 1 over √3. θ = 30° (or π over 6). [1 Mark]

• **[Boxed Result: Acute angle = 30° (π over 6)]**

4. Find the distance of the point (3, -5) from the line 3x - 4y - 26 = 0. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Perpendicular Distance Formula]** d = |A x1 + B y1 + C| over √(A^2 + B^2). Given point (3, -5) and line 3x - 4y - 26 = 0: A = 3, B = -4, C = -26. [1.5 Marks]

• **[Step 2: Calculation]** d = |3(3) - 4(-5) - 26| over √(3^2 + $(-4)^2$) = |9 + 20 - 26| over √(9 + 16) = |3| over √(25) = 3 over 5 = 0.6 units. [1.5 Marks]

• **[Boxed Result: Distance = 3 over 5 units (0.6 units)]**

5. Find the distance between the parallel lines: 3x - 4y + 7 = 0 and 3x - 4y + 5 = 0. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Parallel Lines Distance Formula]** d = |C1 - C2| over √(A^2 + B^2). [1 Mark]

• **[Step 2: Identification & Calculation]** A = 3, B = -4, C1 = 7, C2 = 5. d = |7 - 5| over √(3^2 + $(-4)^2$) = |2| over √(9 + 16) = 2 over √(25) = 2 over 5 = 0.4 units. [2 Marks]

• **[Boxed Result: Distance = 2 over 5 units (0.4 units)]**

6. (a) If p and q are the lengths of perpendiculars from the origin to the lines x cos θ - y sin θ = k cos 2 θ and x sec θ + y cosec θ = k respectively, prove that $p^2 + 4 q^2 = k^2$. (3 Marks)n(b) Find the equation of a line passing through (2, 2) such that the sum of its intercepts on axes is 9. (1 Mark) [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Length p from Origin]** Line 1: x cos θ - y sin θ - k cos 2 θ = 0. p = |0 - 0 - k cos 2 θ| over √($\cos^2$ θ + $\sin^2$ θ) = k cos 2 θ. $p^2 = k^2 \cos^2(2\theta)$. [1.5 Marks]

• **[Step 2: Length q from Origin]** Line 2: x over cos θ + y over sin θ = k ⇒ x sin θ + y cos θ - k sin θ cos θ = 0. q = |0 - k sin θ cos θ| over √($\sin^2$ θ + $\cos^2$ θ) = k sin θ cos θ = (k over 2) sin 2 θ. $4 q^2 = 4 \times (k^2$ over 4) $\sin^2(2\theta) = k^2 \sin^2(2\theta)$. [1.5 Marks]

• **[Step 3: Sum $p^2 + 4 q^2$]** $p^2 + 4 q^2 = k^2 \cos^2(2\theta) + k^2 \sin^2(2\theta) = k^2 (\cos^2 2\theta + \sin^2 2\theta) = k^2$. Proved. (b) Intercepts a + b = 9 and 2 over a + 2 over b = 1 gives a = 6, b = 3 or a = 3, b = 6. Lines: x + 2y - 6 = 0 or 2x + y - 6 = 0. [1 Mark]

• **[Boxed Result: $p^2 + 4 q^2 = k^2$ proved; Lines: x + 2y = 6 or 2x + y = 6]**

7. Find the image of the point (3, 8) with respect to the line x + 3y = 7 assuming the line to be a plane mirror.
[Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Formula for Reflection Image]** For point (x1, y1) and line a x + b y + c = 0: (x - x1) over a = (y - y1) over b = -2 (a x1 + b y1 + c) over (a^2 + b^2). [1.5 Marks]

• **[Step 2: Substitution]** (x1, y1) = (3, 8), line x + 3y - 7 = 0 ⇒ a = 1, b = 3, c = -7. a x1 + b y1 + c = 1(3) + 3(8) - 7 = 3 + 24 - 7 = 20. a^2 + b^2 = 1^2 + 3^2 = 10. Ratio = -2(20) over 10 = -40 over 10 = -4. [1.5 Marks]

• **[Step 3: Calculating Image Coordinates]** (x - 3) over 1 = -4 ⇒ x = 3 - 4 = -1. (y - 8) over 3 = -4 ⇒ y - 8 = -12 ⇒ y = 8 - 12 = -4. Image point = (-1, -4). [1 Mark]

• **[Boxed Result: Image of point = (-1, -4)]**